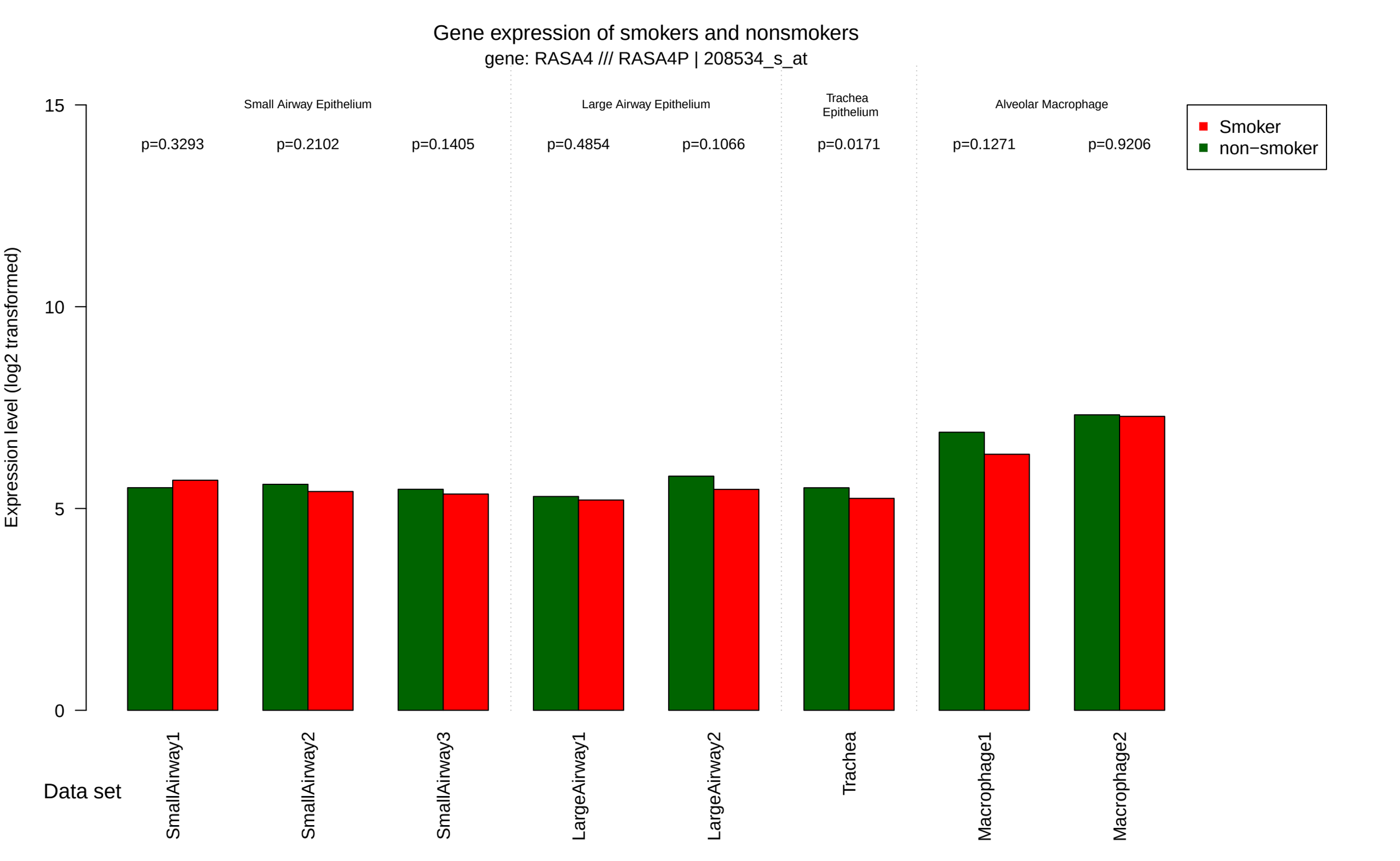
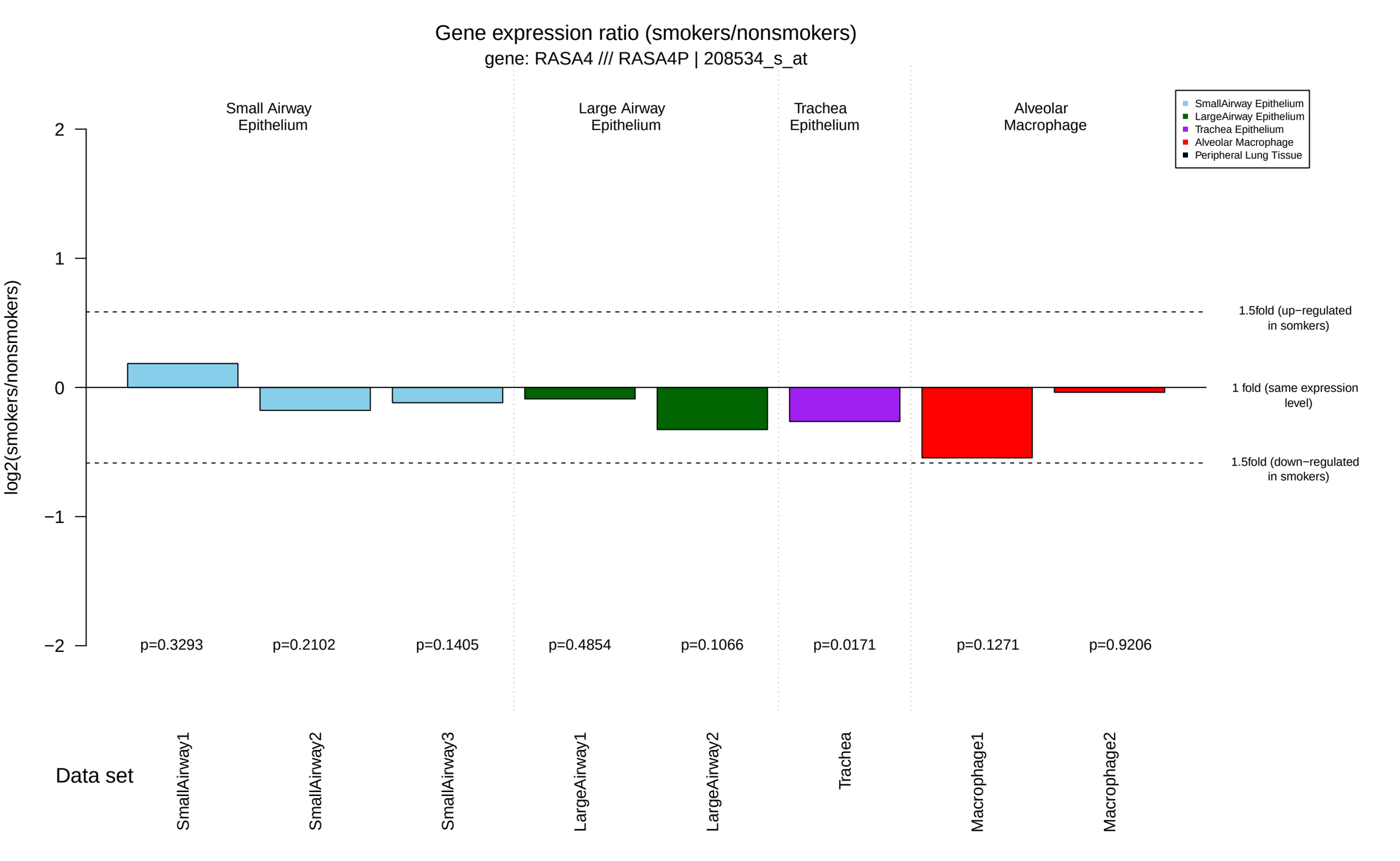
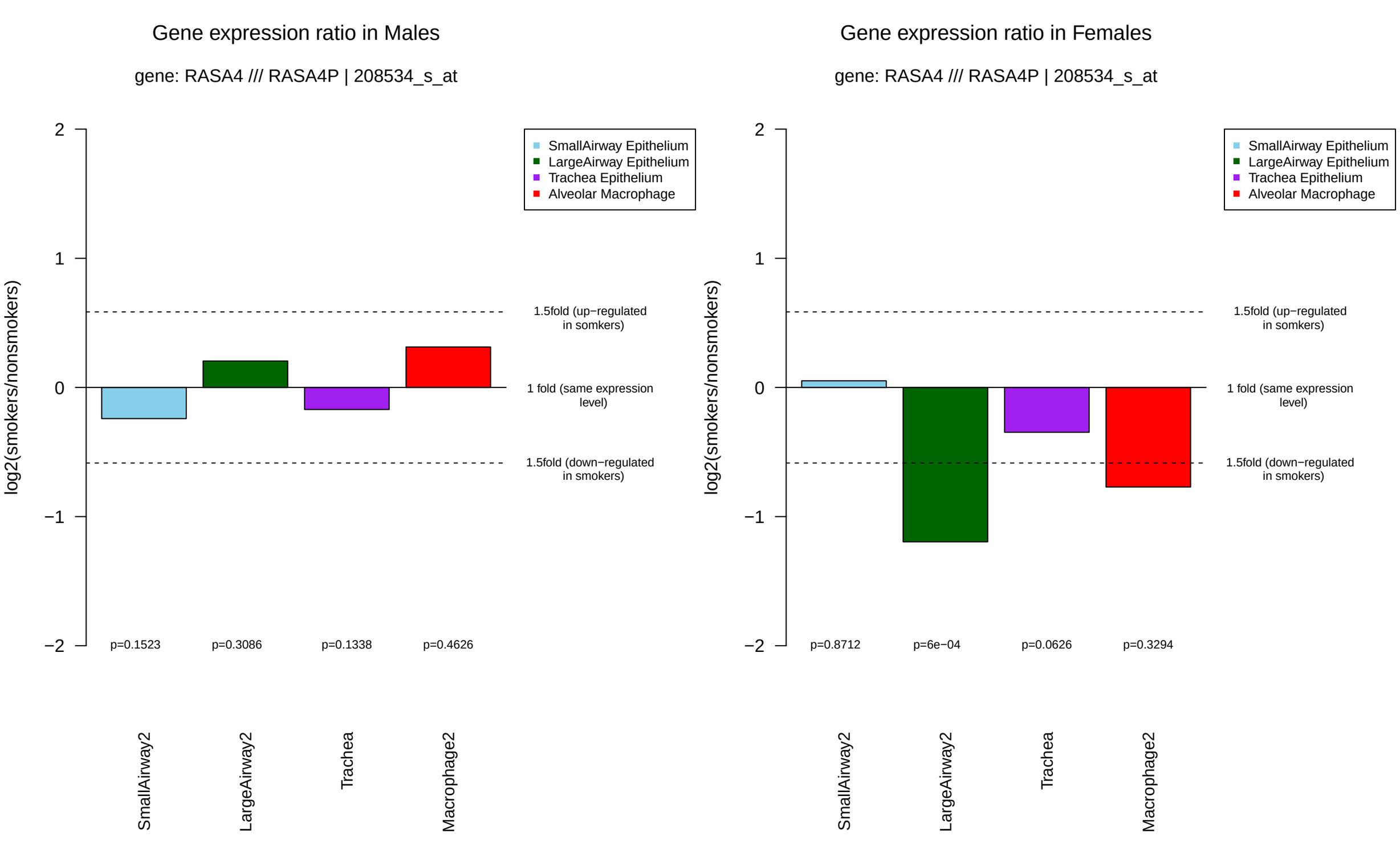
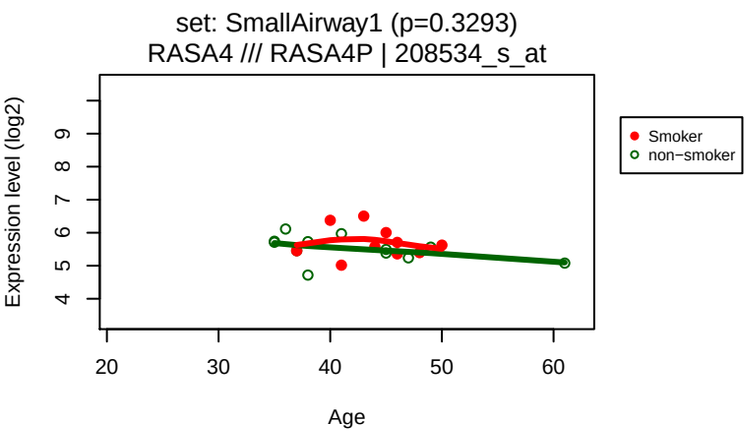
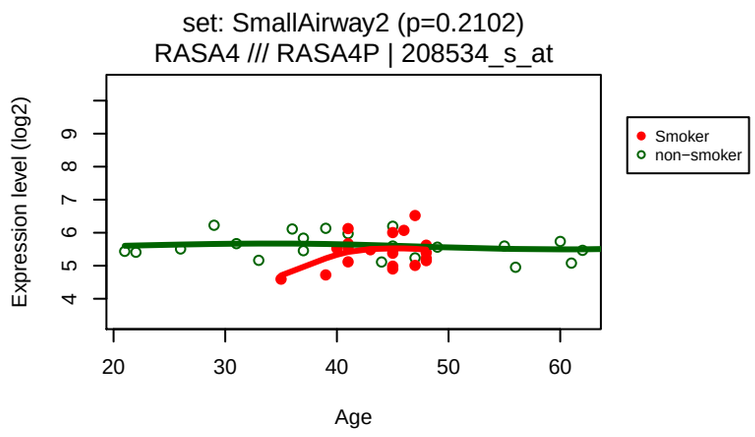
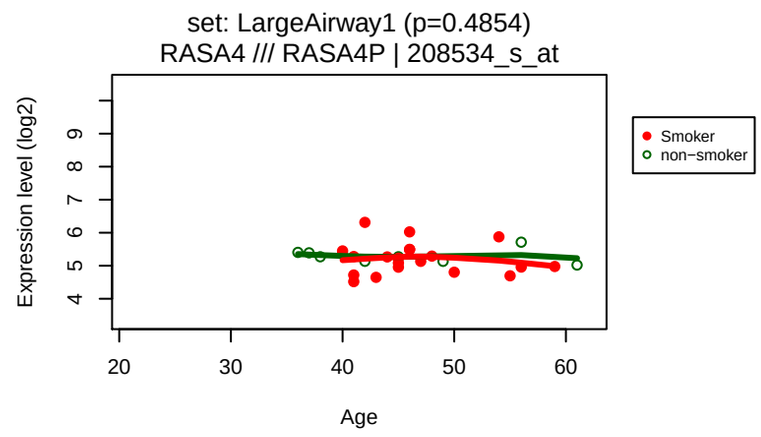

Gene expression of smokers and nonsmokers
gene: RASA4 /// RASA4P | 208534_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3293
p=0.2102
p=0.1405
p=0.4854
p=0.1066
p=0.0171
p=0.1271
p=0.9206
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

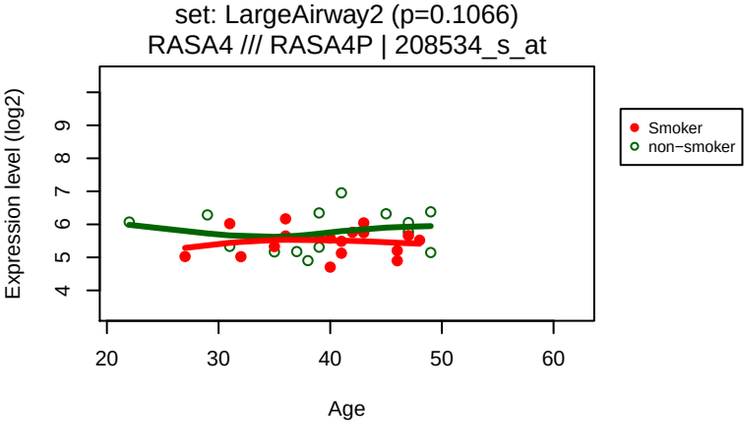

Gene expression ratio (smokers/nonsmokers)
gene: RASA4 /// RASA4P | 208534_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3293
p=0.2102
p=0.1405
p=0.4854
p=0.1066
p=0.0171
p=0.1271
p=0.9206
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

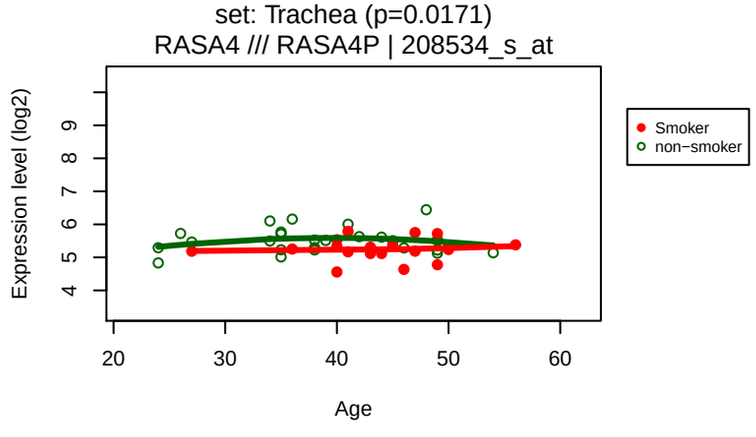

Gene expression ratio in Males
gene: RASA4 /// RASA4P | 208534_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1523
p=0.3086
p=0.1338
p=0.4626
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASA4 /// RASA4P | 208534_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8712
p=6e−04
p=0.0626
p=0.3294
SmallAirway2
LargeAirway2
Trachea
Macrophage2

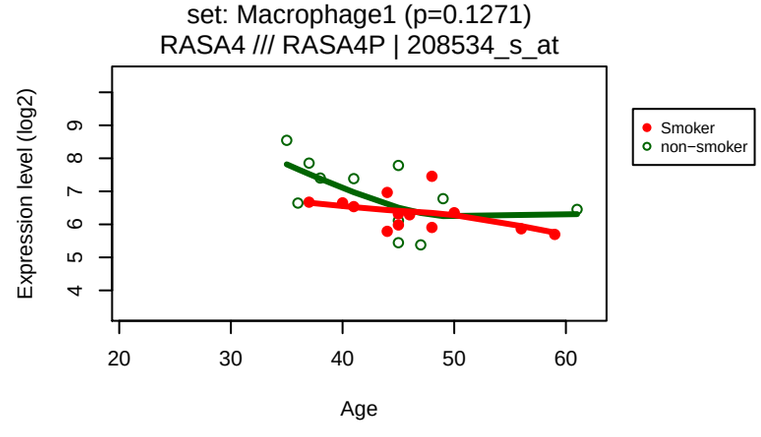

set: SmallAirway1 (p=0.3293)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker

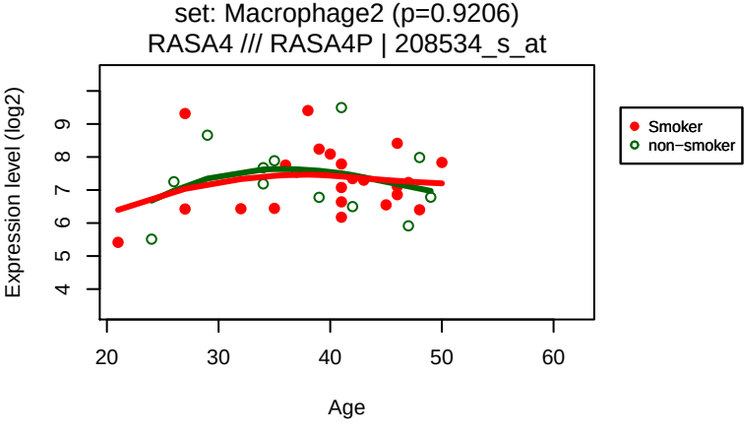

set: SmallAirway2 (p=0.2102)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker

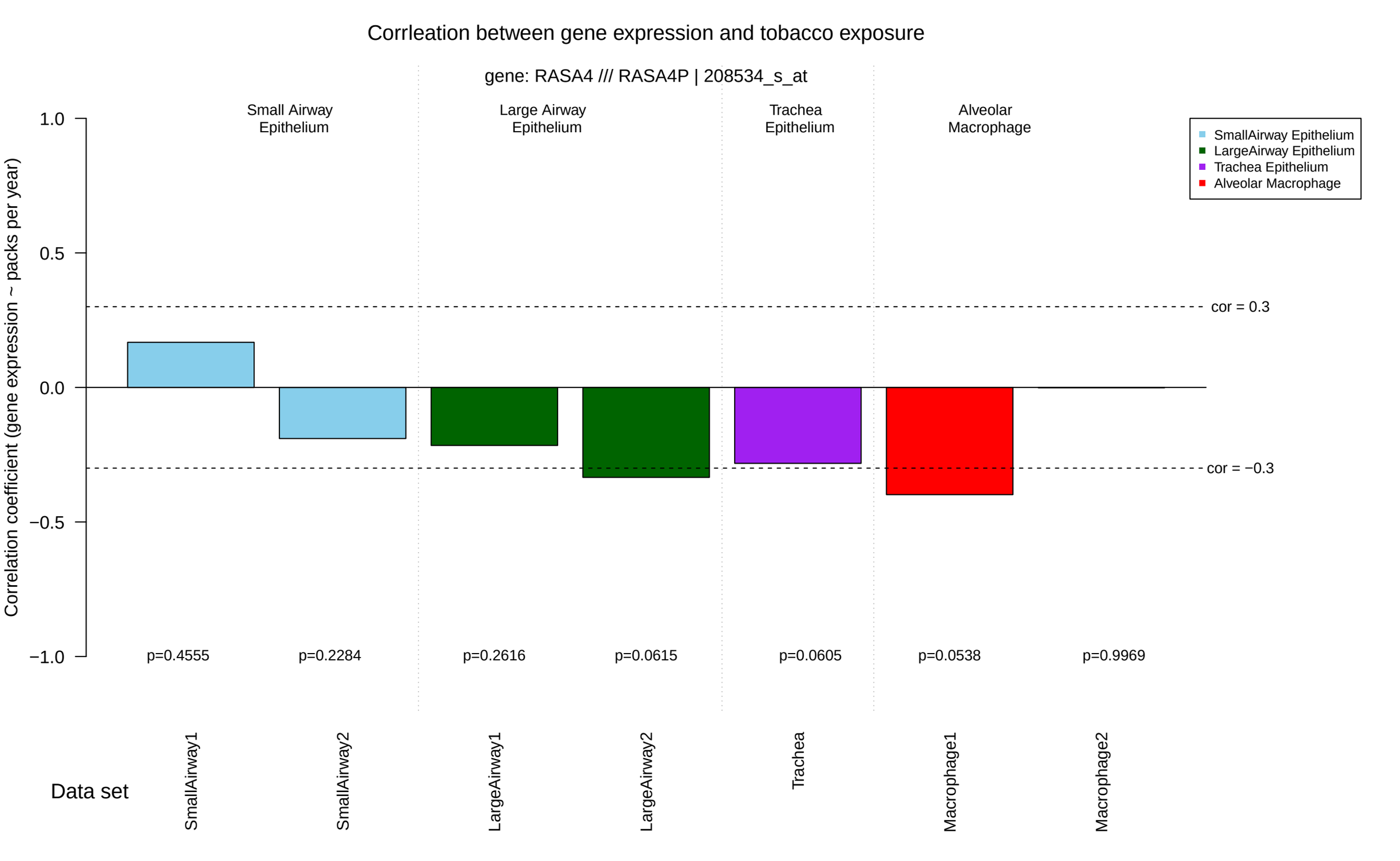

set: LargeAirway1 (p=0.4854)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1066)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0171)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1271)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9206)
RASA4 /// RASA4P | 208534_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASA4 /// RASA4P | 208534_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4555
p=0.2284
p=0.2616
p=0.0615
p=0.0605
p=0.0538
p=0.9969
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2